

Hierarchical Visual Localization Using otgaussvld

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Abstract—Visual localization in large-scale scenes hinges on the quality of global retrieval. This paper presents otgaussvld, an optimal-transport-based Gaussian residual aggregator for place recognition. The proposed layer projects backbone tokens into a latent descriptor space, scores them with diagonal-Gaussian energy or normalized cosine similarity, and solves a Sinkhorn transport problem with a learnable dustbin and token-mass prediction. Transport mass is retained at the cluster level and used to build mean and standard-deviation residuals with respect to learnable Gaussian prototypes. The concatenated 2KD-dimensional descriptor is directly ℓ_2 -normalized without an additional projection head. Experiments on Pittsburgh 30k and Aachen Day-Night demonstrate faster convergence and improved retrieval and localization performance over NetVLAD within a standard hierarchical localization pipeline.

Index Terms—Visual Localization, optimal transport, Gaussian mixture models, Sinkhorn iterations, place recognition.

I. Introduction

Visual localization typically involves 6 Degrees of Freedom (6DoF) positioning of a camera in three-dimensional space, which encompasses three degrees of freedom for three-dimensional translation and three degrees for three-dimensional rotation, corresponding to the three-dimensional position and three Euler angles in the positioning results. To provide positioning results within a consistent reference frame, 6DoF localization relies on a three-dimensional model of the scene, which can be obtained through Simultaneous Localization and Mapping (SLAM) or Structure from Motion (SfM). SLAM focuses more on the real-time aspect of mapping, discarding information and selecting only key frames for map construction during this process. SfM, on the other hand, places greater emphasis on the accuracy of the model, often at the expense of slower mapping speeds. Although both types of models are usable, in environments where real-time mapping requirements are less critical, there is a preference for the more accurate SfM models. In

recent years, visual localization methods based on three-dimensional scene models have primarily been categorized into two approaches: direct 2D-to-3D matching methods and image retrieval-based methods.

Direct 2D-to-3D matching methods involve comparing feature descriptors extracted from query images with those of 3D points in the scene's three-dimensional model to obtain 2D-3D match pairs [1], [2]. Since there may be some incorrect matches among these pairs, a Perspective-n-Point (PnP) algorithm based on Random Sample Consensus (RANSAC) is employed to estimate the camera pose from these 2D-3D matches, thereby filtering out the majority of incorrect matches. These methods typically rely on indexing techniques for matching, which can be computationally expensive, particularly in large-scale scenes with dense descriptor collections. Despite a series of improvements, there remains a trade-off between positioning accuracy and efficiency [3], [4].

Early image-based localization methods treated the localization problem as a place recognition issue, using the pose of the most similar image retrieved from a database as the pose of the query image. DenseVLAD [5] and NetVLAD [6] are representative of such methods. Both are effective in dealing with changes in lighting conditions and perform well in large-scale scenarios, although their positioning accuracy is somewhat lacking. Current image retrieval-based methods typically combine place recognition with indirect 2D-to-3D matching: initially, place recognition is used to narrow down the localization scene, followed by 2D-to-2D matching to establish 2D-to-3D correspondences, leading to refined pose estimation. These methods essentially employ a two-stage process for localization and are thus referred to as hierarchical localization based on image retrieval [7], [8].

Hierarchical localization based on image retrieval has demonstrated notable advantages in scalability, reliability, and effectiveness, establishing itself as a dominant paradigm for large-scale visual localization [8]. However, its performance critically depends on the quality of the global retrieval stage: failure to retrieve semantically coherent local regions leads to significant degradation in pose estimation accuracy. While increasing the number of retrieved images improves recall, it linearly escalates computational costs, hindering real-time applicability. In challenging scenarios such as low-texture areas, viewpoint changes, or illumination variations, the robustness of feature extraction is further compromised. Global retrieval methods like NetVLAD [6] rely solely on first-order statistical aggregation, lacking explicit modeling of local feature distribution differences, which frequently

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causes mismatches under cross-season or cross-view conditions. Meanwhile, local feature extractors face a trade-off between performance and efficiency: models such as SuperPoint [9] and DISK [10] achieve high discriminability at the cost of substantial computational overhead, while lightweight alternatives like XFeat [11] suffer from insufficient salient region perception due to overly simplified network architectures, limiting feature distinctiveness.

To effectively address the aforementioned issues, we have made the following contributions:

(1) We propose otgaussvlad, a global feature aggregation framework that couples diagonal-Gaussian scoring with Sinkhorn transport and a dustbin, thereby producing assignments under explicit mass constraints rather than independent token-wise softmax normalization.

(2) We derive a transport-aware Gaussian residual descriptor that preserves cluster mass, models both mean and scale deviations, and admits a diagonal-Gaussian Wasserstein interpretation of the resulting cluster residuals.

(3) Extensive experiments show that the proposed aggregator improves retrieval accuracy and training stability over NetVLAD while remaining fully compatible with standard hierarchical localization back ends.

II. Proposed Method

A. Framework

The hierarchical visual localization framework consists of two primary stages: an offline mapping phase and an online localization phase, with the overall workflow illustrated in Fig. 1. The objective of this framework is to efficiently and robustly estimate the 3D position and orientation (6-DoF pose) of a camera based on a single query image.

The core objective of the offline mapping phase is to construct a sparse 3D point cloud model (map) of the target scene and precompute essential features. Specifically, we employ an incremental Structure-from-Motion (SfM) algorithm for 3D reconstruction. This process begins by processing a collection of database images using a sparse local feature extractor and the proposed otgaussvlad global descriptor. The extracted local features are used for feature matching and triangulation steps within SfM to recover the 3D structure and the precise pose for each database image. The global descriptors are also computed and stored. The final mapping results, which encompass the 3D point cloud, the database images' poses, their corresponding local features, and global descriptors, are stored in a database to support efficient online localization.

The online localization phase employs a coarse-to-fine pipeline, comprising three sequential steps. First, in the global retrieval stage, the query image is processed by the otgaussvlad method to extract a global feature descriptor. This descriptor enables rapid similarity computation and K-Nearest Neighbor (KNN) search against precomputed database features, yielding a shortlist of visually similar candidate images. Second, during local feature matching

and 2D-3D correspondence establishment, the query and top candidate images are processed by a sparse local feature extractor to obtain local descriptors. Robust 2D-2D feature matching is first established between the query image and candidate database images. These matches are subsequently converted into 2D-3D point correspondences for the query by leveraging the pre-existing associations between 2D keypoints in the candidate images and 3D points in the Structure-from-Motion (SfM) model. Finally, for pose estimation, a RANSAC-based Perspective-n-Point algorithm utilizes the 2D-3D correspondences to estimate the final 6-DoF camera pose.

B. Transport-Aware Gaussian Residual Aggregator

Let $X \in \mathbb{R}^{C \times H \times W}$ denote the backbone output. After the two-layer point-wise adapter and LayerNorm, the feature map is flattened into $N = HW$ local tokens $z_n \in \mathbb{R}^D$. In the default configuration, $C = 768$, $D = 64$, and $K = 64$, resulting in a raw descriptor dimension of $2KD = 8192$. No projection head is applied after aggregation.

For each cluster $k \in \{1, \dots, K\}$, the layer parameterizes a diagonal Gaussian prototype

$$p_k(z) = \mathcal{N}(z; \mu_k, \text{diag}(\sigma_k^2)),$$

where $\mu_k \in \mathbb{R}^D$, $\sigma_k \in \mathbb{R}_+^D$, and the mixture prior is

$$\pi_k = \frac{e^{\alpha_k}}{\sum_{j=1}^K e^{\alpha_j}}.$$

The default GMM score is the corresponding log-density up to a cluster-independent constant:

$$s_{kn} = \log \pi_k - \frac{1}{2} \sum_{d=1}^D \frac{(z_{nd} - \mu_{kd})^2}{\sigma_{kd}^2} - \sum_{d=1}^D \log \sigma_{kd}. \quad (1)$$

For completeness, the attention ablation uses normalized prototypes and tokens, $\bar{\mu}_k = \mu_k / \|\mu_k\|_2$ and $\bar{z}_n = z_n / \|z_n\|_2$, with score $s_{kn}^{\text{att}} = \tau \langle \bar{\mu}_k, \bar{z}_n \rangle$, where $\tau = \text{softplus}(a) + \epsilon$.

When token_mass_mode is set to learned, a lightweight token saliency head produces logits q_n and masses

$$\nu_n = \frac{\exp(\eta q_n)}{\sum_{m=1}^N \exp(\eta q_m)}, \quad \eta = \text{softplus}(\lambda) + \epsilon.$$

In the uniform case, $\nu_n = 1/N$. The target mass vector is $n = \nu / \|\nu\|_1$.

To suppress uninformative tokens, the score matrix is augmented with a learnable dustbin row and the following entropy-regularized transport problem is solved:

$$A^* = \arg \max_{A \in \mathcal{U}(m, n)} \langle \tilde{S}, A \rangle + \mathcal{H}(A),$$

where $\mathcal{U}(m, n) = \{A \geq 0 : A\mathbf{1} = m, A^\top \mathbf{1} = n\}$, $\mathcal{H}(A) = -\sum_{k,n} A_{kn} \log A_{kn}$, and

$$m = [(1 - \delta)\pi_1, \dots, (1 - \delta)\pi_K, \delta]^\top.$$

Here $\delta \in (0, 1)$ is the dustbin mass and $\tilde{S}$ denotes the score matrix after appending the dustbin row. The implemented

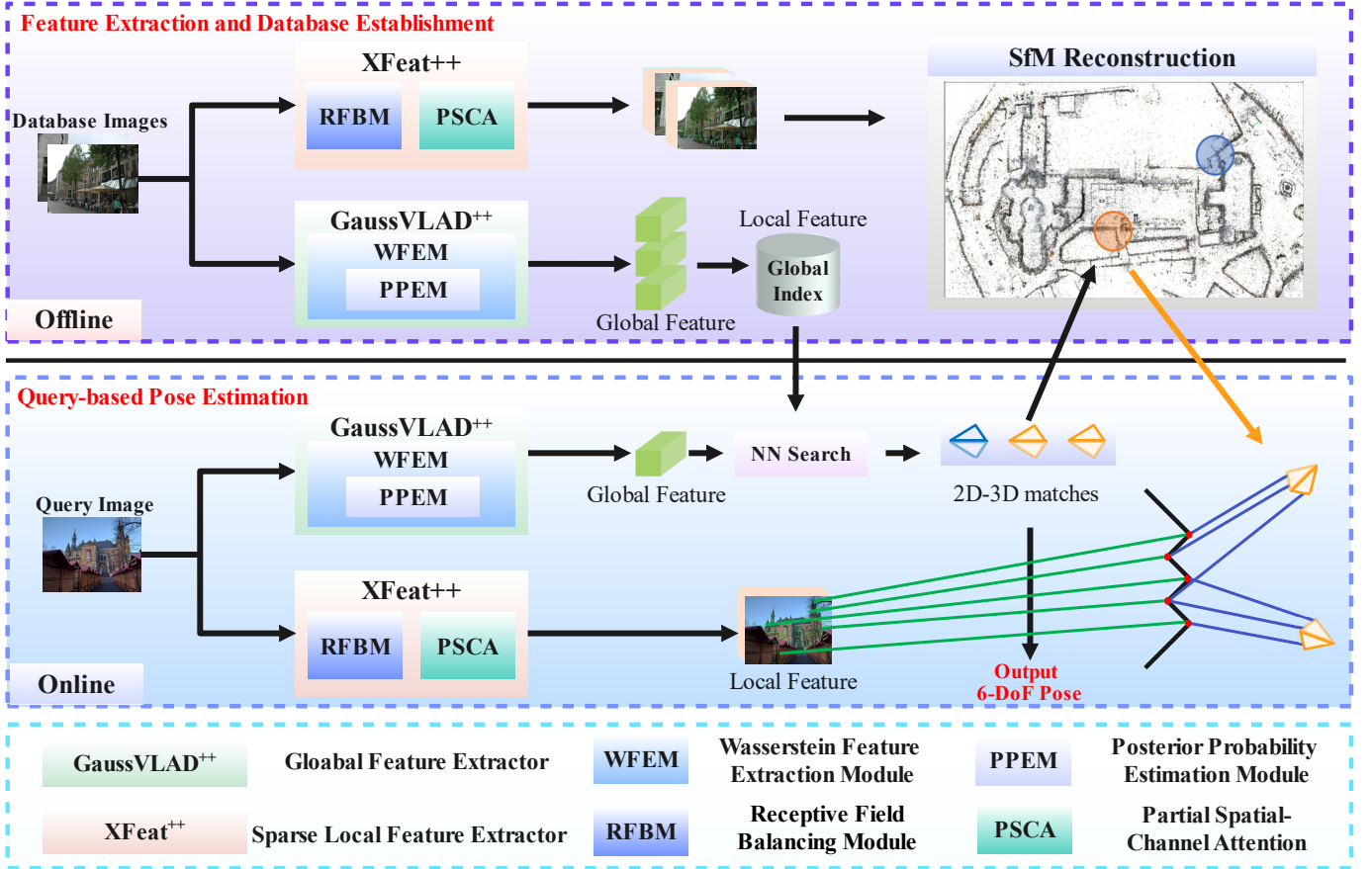


Fig. 1. Flowchart of the hierarchical visual localization system. The system is divided into two phases: offline and online. In the offline phase, database images are processed by a sparse local feature extractor and otgaussvlad to construct a database containing a global index and a sparse 3D point cloud (SfM Reconstruction). In the online phase, a query image is first processed by otgaussvlad for global retrieval to identify candidate scene regions. Subsequently, sparse local features are matched with the retrieved candidates in a 2D-3D matching step. Finally, the precise 6-DoF camera pose is estimated using a PnP algorithm.

log-space Sinkhorn iterations compute the associated dual scalings.

Proposition 1 (Well-posedness of the transport assignment). If the source mass m and target mass n are strictly positive, then the entropy-regularized transport problem admits at least one maximizer. Because the entropy term is strictly concave on the interior of $\mathcal{U}(m, n)$, the maximizer is unique.

Proof. The feasible set $\mathcal{U}(m, n)$ is non-empty, closed, and bounded, hence compact. The map $A \mapsto \langle \tilde{S}, A \rangle$ is linear, and $\mathcal{H}(A)$ is strictly concave on the positive orthant. Therefore the objective is strictly concave over a compact convex set, which yields existence and uniqueness of the maximizer. $\square$

Let A be the real-cluster block of A^* . For each cluster k , define the transported mass

$$m_k = \sum_{n=1}^N A_{kn}, \quad \gamma_{kn} = \frac{A_{kn}}{\max(m_k, \epsilon)}.$$

If $m_k > 0$, then $\sum_n \gamma_{kn} = 1$, so $\gamma_{k\cdot}$ defines a probability distribution over the tokens assigned to cluster k . The

transport-induced empirical moments are then

$$\hat{\mu}_k = \sum_{n=1}^N \gamma_{kn} z_n, \quad (2)$$

$$\hat{v}_k = \sum_{n=1}^N \gamma_{kn} (z_n - \hat{\mu}_k) \odot (z_n - \hat{\mu}_k), \quad (3)$$

$$\hat{\sigma}_k = \sqrt{\hat{v}_k}. \quad (4)$$

Proposition 2 (Transport-induced Gaussian moments). The quantities $\hat{\mu}_k$ and $\hat{v}_k$ are exactly the first and diagonal second central moments of the empirical distribution $q_k = \sum_n \gamma_{kn} \delta_{z_n}$. Equivalently, $\hat{\mu}_k$ is the unique minimizer of $\sum_n \gamma_{kn} \|z_n - \mu\|_2^2$.

Proof. The normalization $\sum_n \gamma_{kn} = 1$ makes $\gamma_{k\cdot}$ a valid discrete probability measure. The first equation is therefore its weighted mean, and the second is the corresponding diagonal central second moment. The minimizer statement follows from standard least-squares optimality of the weighted mean. $\square$

The cluster residual is formed as

$$r_k = [\hat{\mu}_k - \mu_k, \hat{\sigma}_k - \sigma_k].$$

This residual has a closed-form Wasserstein interpretation under diagonal Gaussian geometry, and its concatenation over clusters serves as the global retrieval feature.

When `mass_preserving` is enabled, the residual is reweighted by

$$w_k = \left(\frac{K}{1 - \delta} m_k \right)^\beta, \quad \beta = \text{mass_power},$$

so that the final cluster residual becomes $\tilde{r}_k = w_k r_k$. This normalization is anchored at the uniform reference mass: if $m_k = (1 - \delta)/K$, then $w_k = 1$. The final global descriptor is obtained by concatenating $\tilde{r}_1, \dots, \tilde{r}_K$ and applying ℓ_2 normalization:

$$g = \frac{\text{vec}([\tilde{r}_1, \dots, \tilde{r}_K])}{\|\text{vec}([\tilde{r}_1, \dots, \tilde{r}_K])\|_2}.$$

Therefore, the layer can be viewed as a prototype-indexed entropy-regularized transport operator followed by a Wasserstein residual encoder.

III. Experimental Results and Analysis

A. Experiment Setup

This section details the implementation specifics and experimental configurations for the proposed `otgaussvld` method. The setup encompasses training datasets, parameter configurations, and hardware/software environments to ensure reproducibility.

1) `otgaussvld`: The training dataset, Pittsburgh 30k [6], comprises approximately 30,000 geotagged street-view images from Google Street View, with GPS coordinates accurate within 7–15 meters. Diversity in capture times, viewing angles, and lighting conditions offers a wide range of positive and hard negative samples. Experiments were conducted on a system equipped with an NVIDIA GeForce RTX 4080 GPU and an Intel Core i9-14900K CPU. The software environment included CUDA 12.4, NVIDIA driver version 550.54.14, PyTorch 1.13.0 with CUDA 11.7 support, and torchvision 0.14.0. In the `otgaussvld` configuration used in our experiments, the aggregator is instantiated with 768 input channels, 64 clusters, a 64-dimensional cluster space, 768 hidden channels, GMM scoring, three Sinkhorn iterations, a dustbin mass of 0.05, learned token mass prediction, mass-preserving residuals, and `mass_power` set to 1.0. This configuration yields an 8192-dimensional global descriptor. The training hyperparameters for both NetVLAD and the proposed `otgaussvld` method were identical: an L2 regularization coefficient of 0.001, initial learning rate of 0.0001, learning rate decay rate of 0.5 applied every 5 epochs, momentum of 0.9, and triplet margin threshold of 0.3.

B. `otgaussvld`

Metrics. $\text{Recall}@K$ is a core metric for global image retrieval, measuring the proportion of correctly matched images among the top- K database results. It evaluates feature encoding efficacy: high $\text{Recall}@1$ reflects top-tier precision, while $\text{Recall}@5$ and higher thresholds indicate

TABLE I
Comparison of Recall Rates on the Test Set

Method	Pittsburgh 250k-test			
	Recall@1	Recall@5	Recall@10	Recall@20
SPE-NetVLAD [14]	85.60%	93.20%	94.55%	95.90%
NetVLAD [6]	81.64%	90.98%	93.26%	95.05%
MR-NetVLAD [17]	86.70%	93.60%	94.80%	96.00%
Ours(VGG)	84.42	92.74	94.87%	96.20%
Ours(EfficientNet)	88.42%	95.91%	97.39%	98.31%

robustness to challenges like illumination or viewpoint variations. This ensures reliable candidate image selection for subsequent local matching and camera pose estimation.

Results. Fig. 2 demonstrates `otgaussvld`'s significant performance advantage over NetVLAD. Initially, `otgaussvld` achieves a lower loss (0.05 vs. NetVLAD's 0.23), indicating superior initialization. It converges rapidly, stabilizing near 0.01 after merely 5 epochs and approaching zero by epoch 15. In contrast, NetVLAD exhibits slow convergence, with loss remaining at 0.17 even after 15 epochs. The results confirm `otgaussvld`'s excellence in training efficiency, model performance, and stability.

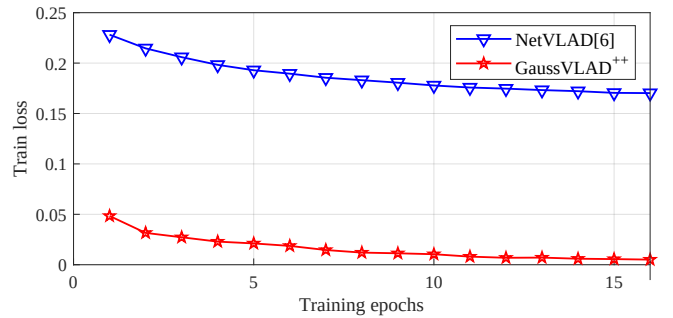


Fig. 2. Comparison of training loss over epochs. The graph shows the change in loss during the training process for NetVLAD and `otgaussvld` methods. The x-axis represents the number of training epochs, and the y-axis represents the training loss.

The comparative evaluation of recall rates on the Pittsburgh 250k-test dataset is summarized in Table I. The proposed method with an EfficientNet backbone achieves state-of-the-art performance across all evaluation metrics, attaining recall rates of 88.42% (Recall@1), 95.91% (Recall@5), 97.39% (Recall@10), and 98.31% (Recall@20). This represents a significant improvement over existing approaches, outperforming the second-best method (MR-NetVLAD) by 1.72% in Recall@1 and by 2.31% in Recall@5.

The comparison highlights the critical role of backbone architecture selection: the EfficientNet variant exhibits a substantial gain over the VGG-based implementation, particularly in Recall@1 (+4.00%) and Recall@5 (+3.17%). Both variants consistently surpass the NetVLAD baseline, validating the efficacy of the proposed feature encoding framework in large-scale visual localization tasks.

To eliminate the impact of the training framework and backbone network on model performance, we uniformly used the VGG network as the Backbone and retrained

NetVLAD and otgaussvlad using PyTorch. To verify the effectiveness of our proposed global retrieval method in localization, we evaluated the role of our method otgaussvlad in localization using the hierarchical localization toolbox (Hloc) under fixed sparse local features and feature matching methods. As shown in Table II, the results indicate that our method can recall database images highly relevant to the query images for both daytime and nighttime scenes. This outcome aligns perfectly with the essence of our design for otgaussvlad, which employs the Wasserstein distance as a measure of feature similarity. This approach better considers the distribution of local semantic features, giving the extracted global features a significant advantage when retrieving candidate images.

TABLE II

Performance Comparison of Global Feature extracting Methods on Aachen Day-night

Method	Day			Night		
	0.25m 2°	0.5m 5°	5m 10°	0.25m 2°	0.5m 5°	5m 10°
NetVLAD [6]	81.6	87.4	92.0	71.4	81.6	91.8
otgaussvlad	83.1	89.9	93.9	79.6	87.8	94.9

IV. Conclusion

This paper presents otgaussvlad, a transport-aware global feature aggregator for hierarchical visual localization. By combining diagonal-Gaussian scoring, Sinkhorn transport with a dustbin, learned token mass, and mass-preserving mean/scale residuals, the proposed layer produces a compact descriptor that is more robust to appearance variation than NetVLAD while remaining fully compatible with conventional retrieval-and-matching pipelines. Experiments on Pittsburgh 30k and Aachen Day-Night confirm faster convergence and improved retrieval and localization performance.

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